

PRJ_409: AI Personal Finance Advisor

System Modules & Primary Methods / Algorithms Used

Module / Component	Primary Method & Algorithm Used
1. Transaction Categorization	TF-IDF Vectorization + Logistic Regression Classifier (with Rule-Based regex fallback)
2. Budget Recommendation Engine	Rolling Historical Moving Average
3. Goal Feasibility Engine	Amortized Capital Allocation & Liquidity Constraint Verification
4. Recurring Cost & Anomaly Detector	Cadence Clustering & Percentage Drift Scoring
5. Affordability Calculator	Multi-Factor Liquidity & Capital Preservation Analysis
6. AI Advisor & Chatbot	23-Intent NLP Classifier with Dynamic State-Bound Formula Simulation
7. Financial Rule-Based Insights	Heuristic Statistical Analysis (Pareto 80/20 & Velocity Growth)
8. Retirement & Inflation Projections	Compounded Future Value & Trinity 4% Safe Withdrawal Rule
9. Authentication & Security	OAuth2 Password Bearer Flow + JWT (HMAC-SHA256) + Argon2/Bcrypt